

Review of generative models for the narrowband land mobile satellite propagation channel

F. P. Fontan^{1,*}, A. Mayo¹, D. Marote¹, R. Prieto-Cerdeira², P. Mariño¹,
F. Machado¹ and N. Riera³

¹*ETS Ing. de Telecomunicación, Universidad de Vigo, Campus Universitario, E-36200 Vigo, Spain*

²*European Space Agency, ESA-ESTEC, TEC-EEP, Keplerlaan 1, 2201 AZ Noordwijk, The Netherlands*

³*German Aerospace Center (DLR), Institute of Communications and Navigation, Oberpfaffenhofen,
82234 Wessling, Germany*

SUMMARY

The land mobile satellite (LMS) propagation channel is frequently described using statistical models. These models usually make different assumptions regarding the behavior of the direct signal, the diffuse multipath component and the shadowing effects. This paper analyzes the theoretical formulation and implementation of time-series synthesizers based on three typical statistical models: Loo, Corazza–Vatalaro and Suzuki, describing their similarities and differences. The discussion is not limited to the amplitude of the complex envelope but also to the phase variations and Doppler spectra. Finally, guidelines are also provided for comparing model parameters supplied by different authors. Copyright © 2008 John Wiley & Sons, Ltd.

KEY WORDS: generative models; statistical models; land mobile satellite (LMS); narrowband propagation channel

1. INTRODUCTION

Performance studies of candidate proposals for new land mobile satellite, LMS, systems for broadcast and two-way communication applications rely on the use of synthetic time-series describing the dynamics of the propagation channel. A number of statistical models exist that try to reproduce the channel behavior for a number of propagation scenarios, e.g. open, tree-shadowed, suburban, urban, etc. Models such as those of Loo [1], Corazza and Vatalaro (C&V) [2] or Suzuki [3] may be considered typical examples that are often used for describing the

*Correspondence to: F. P. Fontan, ETS Ing. de Telecomunicación, Universidad de Vigo, Campus Universitario, E-36200 Vigo, Spain.

†E-mail: fpfontan@tsc.uvigo.es

dynamics of the narrowband LMS channel both for geostationary orbit (GEO) and non-GEO satellite-based systems. This paper deals with time-series generators based on those statistical models.

Two main elements are considered in narrowband channel modeling, namely, the *direct signal* and *diffuse multipath*. Assumptions are made as to the statistical behavior of these two components. Typical assumptions such as lognormal direct signal variations due to shadowing plus a constant multipath power or constant direct signal plus multipath multiplied by lognormal shadowing variations are quite common, and their associated *complementary cumulative distribution functions*, CCDFs, have been shown to agree fairly well with experimental data statistics [1,2,4].

One basic application of statistical models is generating time-series for dynamic system performance evaluation. Statistical models, conversely to what happens for purely empirical models, are very well suited for generating time-series. Statistical models make a number of physical assumptions on the behavior of the channel's main elements: the direct signal and the (diffuse) multipath component. However, their parameters must be extracted from experimental data. Thus, statistical models represent an intermediate stage between purely empirical models and those of physical or deterministic nature.

In this paper, attention is paid to the narrowband channel case, which is the most probable condition found in practice in the LMS channel [4]. Events in which the channel gives rise to large time dispersion are seldom found, and correspond to spots along the mobile terminal route clear of near obstacles so that far echoes with long excess delays can reach the receiver from large terrain and building features (especially specular rays). Through geometrical considerations, it may be concluded that far echoes will more likely occur at low satellite elevations, while, for the higher elevations, Snell's law of reflection will not be fulfilled except for very high obstacles.

The term *time-series* is used here (*narrowband case*) to refer to the variations in the complex envelope of the received signal, i.e. both amplitude and phase. When *wideband conditions* occur, the term time-series would refer to the complex series for the different taps in a tapped delay-line channel model. When a narrowband channel is simulated, the transmitted signal is multiplied by the synthesized complex-envelope variations that encompass both the shadowing, fading and Doppler phenomena.

Another feature of interest is the *rate of change* of the different channel components involved: shadowing and multipath. *Multipath fading* is produced as a consequence of ray interference, and gives rise to signal enhancements and cancellations taking place in a fraction of the wavelength: a maximum of two peaks and two troughs can be observed within one wavelength. On the other hand, *shadowing* is a slower process, which is directly related to the sizes of the environmental features in the near vicinity of the terminal.

Furthermore, in the LMS channel, shadowing effects may show significant changes in received level due to large environmental features, e.g. received levels at a moving terminal alternating line-of-sight (LOS) and blocked conditions. Typically, these pronounced, larger-scale variations can be modeled into different 'states' (e.g. using Markov chains), i.e. the LOS and non-LOS states. Sometimes, it is best to define the so-called GOOD and BAD states that refer to LOS or light shadowing conditions, and moderate to heavy shadowing conditions, respectively.

On the other hand, the shadowing can also induce smaller-scale signal level changes around a given mean value. For instance, this occurs when the terminal traverses a non-uniform tree-shadowed area, i.e. different branch and leaf densities.

Combined statistical distributions such as those of Loo, C&V or Suzuki [1–4] agree very well with measured data when the shadowing conditions do not change drastically, i.e. when the received signal remains in the same ‘state’. However, when used to model situations where both LOS conditions and deep shadowing conditions are found, then the agreement may not be very good.

This is illustrated in Figures 1 and 2. In Figure 1 a time-series measured in a tree-shadowed area [4] is shown together with its corresponding *probability density function* (pdf). In Figure 2 the overall CCDF corresponding to the time-series in Figure 1 is shown. It can be observed how the CCDF curve (or its corresponding pdf) may be split into two distinct regions corresponding to the blocked and non-blocked conditions. This CCDF cannot be described by means of a single theoretical distribution: note the inflection point clearly indicating two differentiated behaviors. However, if the total time-series is split into homogeneous (same state) segments, good fits may be obtained between measured and theoretical distributions.

Thus, a *two-stage approach* in the description of shadowing effects is often assumed, as described in Figure 3. In the figure, three different shadowing/fading rates are sketched:

- 1 very slow variations,
- 2 slow variations,
- 3 fast variations.

The *very slow variations* correspond to shadowing events caused by single large features or grouped features: trees, houses, etc., which can be mapped to a single Markov state. The *slow variations* are caused by the non-uniformity of single large obstacles; for instance, different branch densities in a group of trees or different building heights in a street. The *fast variations* are caused by multipath.

These observations clearly suggest that combinations of distributions such those proposed by Loo, C&V or Suzuki cannot be used directly for characterizing the overall channel dynamics: GOOD and BAD states together. Rather, a combination of distributions is required. The overall pdf in such cases will follow the general expression given below where GOOD and BAD shadowing sections are found along the terminal’s route:

$$f_{\text{Overall}}(r) = p_{\text{GOOD}} \times f_{\text{GOOD}}(r) + p_{\text{BAD}} \times f_{\text{BAD}}(r) \quad (1)$$

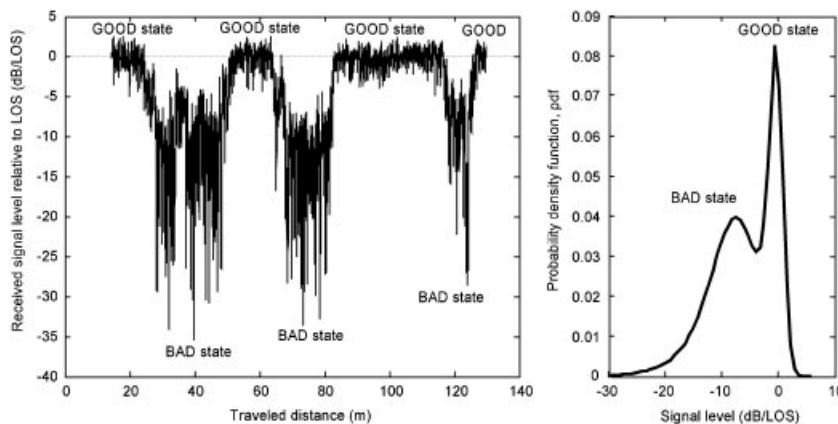


Figure 1. S-band measured time-series in a tree-shadowed environment and 40° elevation [4].

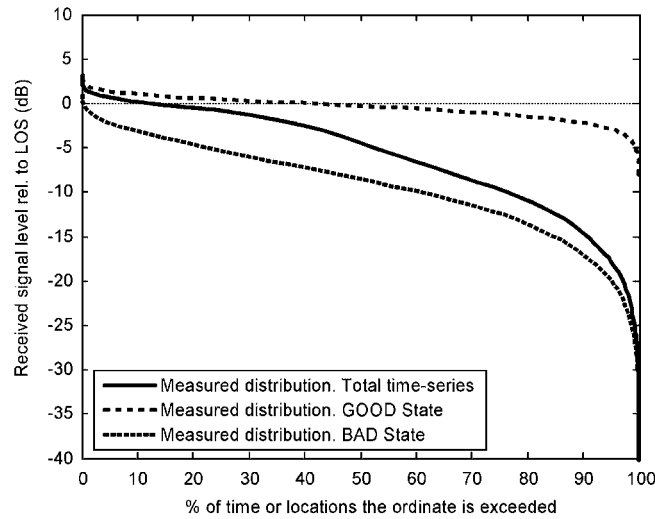


Figure 2. Measured distributions: overall, GOOD and BAD states, corresponding to the series in Figure 1.

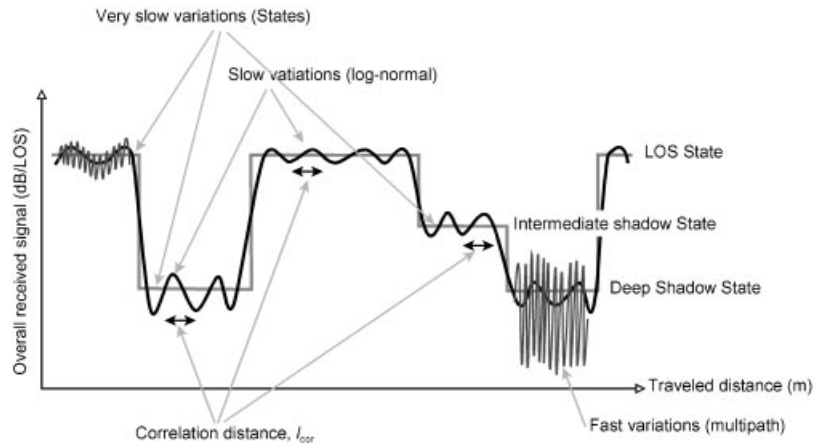


Figure 3. Markov model and different rates of change in the narrowband received amplitude within each state.

with p_{GOOD} the probability that the link is in the GOOD shadowing conditions, f_{GOOD} the pdf of the amplitude variations (r) when in the GOOD condition, p_{BAD} the probability of the link being under BAD shadowing conditions and f_{BAD} the pdf of the signal variations when in the BAD condition.

This paper deals only with those statistical models that describe *the signal variations within one of the possible 'states'* (Figure 3). Those models still have to account for multipath and smaller-scale shadowing effects. This means that f_{GOOD} and f_{BAD} result from the combination of a distribution describing multipath and another describing the smaller range shadowing

effects. For an overall description of a Markov chain-based model see Reference [5]. In that paper, a summary of parameters for L- and S-band is provided including Loo distribution parameters and state transition probabilities.

One final aspect is that of the spectral shaping of the multipath contribution. For reproducing the Doppler effects, the U-shaped, Jakes spectrum [6] model is often used. A more realistic approach seems to be one assuming a Butterworth model according to [7]. However, in this paper simulations are only performed using the classical Jakes spectrum. To this spectrum, a delta for the direct component must be added. Spectral shaping is achieved through the coloring (filtering) of two white Gaussian noise sources in quadrature.

2. LMS STATISTICAL CHANNEL MODELS

In the cellular/terrestrial channel, multipath effects are best described by means of the Rayleigh distribution, which assumes that the direct signal is totally blocked. However, for the LMS channel, the most commonly used distribution is that of Rice, which assumes that the direct signal may be attenuated but is not negligible. In any event, the Rice distribution encompasses the Rayleigh distribution as a special case when the direct signal is equal to zero.

The Rice distribution is used due to the fact that LMS systems need to operate with small fade margins and the direct signal must be present and unshadowed. However, recently, with the on-going interest and the standardization activities underway in the field of direct satellite to mobile TV broadcast systems, Satellite Services for Portable devices (DVB-SSP) [8], it is expected that larger fade margins than those of earlier systems (Globalstar, Iridium, etc.) will be available. Thus, it is interesting to revisit the existing models and refine their parameters, especially for heavier shadowing conditions [5].

In spite of the expected fade margin increase, still, the direct signal must be either completely unblocked or only moderately shadowed. Purely Rayleigh conditions, implying that the direct signal is totally lost, are of little use unless the amount of diffuse multipath power is large enough.

Normally, to make LMS communications or broadcast systems feasible, multi-satellite diversity techniques will be required to achieve acceptable levels of availability. In broadcast systems, terrestrial gap-fillers could provide the required availability within urban areas, whereas satellite diversity would be used in highways or suburban areas.

Here, we will be assuming signals normalized with respect to the free space (FS) level. Thus, $r(t)$ or $r(x)$ represents the normalized complex envelope at a given point in time, t , or location, x . The normalized voltage, received under FS conditions, will be set such that $|r(t)| = |r(x)| = 1$ or, in dB, $20 \log |r(t)| = 20 \log |r(x)| = 0$ dB.

The actual received power and voltage are linked through $p = v^2/(2R)$ where R is the load resistance. Normalizing with respect to FS received power yields

$$p' = \frac{p}{p_{FS}} = \frac{v^2/2R}{v_{FS}^2/2R} = \frac{v^2}{v_{FS}^2} = r^2 \quad (2)$$

Variable p' is the normalized power and r , actually $|r|$, the normalized voltage. Thus, if the normalized voltage is, for example, $r = 0.3$, this is equivalent to a level of $20 \log(0.3) = -10.46$ dB with respect to the FS level.

The *Rice distribution* depends on two parameters, the normalized direct ray amplitude, a (fixed vector), and the normalized average power of the multipath component (random vector), $2\sigma^2$, or, in dB, $MP = 10 \log(2\sigma^2)$, where MP stands for *multipath power*. As said, if $a = 0$, the *Rice* distribution becomes a Rayleigh distribution.

Depending on the assumptions, two interpretations can be made of the Rice distribution [9]:

- *Assumption (a)*: The power of the fixed vector is constant while the overall power of the combination of the fixed and random vectors varies. In this case, the average overall power is given by the expression

$$a^2 + 2\sigma^2 = a^2(1 + k)/k \quad (3)$$

where $k = a^2/2\sigma^2$ is the fixed-to-random vector power (carrier-to-multipath) ratio.

- *Assumption (b)*: The overall power (fixed plus random vector) remains constant, but the powers of both components vary.

In the two cases, the shapes of their *cumulative distribution functions* (CDFs) are identical except for an offset in the signal level axis (expressed in dB) of

$$10 \log\left(\frac{k+1}{k}\right) \text{ dB} \quad (4)$$

The *Loo distribution* [1] is based on the Rice distribution. In this case, assumption (a) is made, i.e. the direct signal is constant while the overall signal power varies. The received signal is supposed to follow a Rice distribution with parameters a and $2\sigma^2$, i.e.

$$f(r|a) = \frac{r}{\sigma^2} \exp\left[-\frac{r^2 + a^2}{2\sigma^2}\right] I_0\left(\frac{ra}{\sigma^2}\right) \quad (5)$$

with $r \geq 0$. This is valid for short sections of the traveled route. For longer sections, the direct signal amplitude is assumed to vary according to a lognormal distribution, i.e.

$$f(a) = \frac{1}{\Sigma_{\text{Np}} a \sqrt{2\pi}} \exp\left[-\frac{(\ln(a) - M_{\text{Np}})^2}{2\Sigma_{\text{Np}}^2}\right] = \frac{8.686}{\Sigma_{\text{dB}} a \sqrt{2\pi}} \exp\left[-\frac{(20 \log(a) - M_{\text{dB}})^2}{2\Sigma_{\text{dB}}^2}\right] \quad (6)$$

with $r \geq 0$, M and Σ are the mean and the standard deviation of its associated normal distribution for the direct signal's amplitude either in Nepers ($\ln$) or in dB ($20 \log$), where the expression $8.686 \ln(x) = 20 \log(x)$ is used for unit conversion. Thus, the conversion of parameters in Nepers and dB is straightforward, i.e. $M_{\text{dB}} = 8.686 M_{\text{Np}}$ and $\Sigma_{\text{dB}} = 8.686 \Sigma_{\text{Np}}$.

Moreover, the Loo model assumes that the received complex envelope consists of the sum of two phasors, i.e.

$$r_T \exp(j\phi_T) = r_D \exp(j\phi_D) + r_M \exp(j\phi_M) \quad (7)$$

where subscript T indicates total or overall, D direct signal and M multipath. As indicated above, the direct signal amplitude is lognormally distributed due to shadowing, while the multipath component parameter, $2\sigma^2$, remains constant.

The overall distribution i.e. for longer, homogeneous (same state) sections of mobile route is given by

$$f(r) = \int_0^\infty f(r|a)f(a) da$$

$$= \frac{8.686r}{\sigma^2 \Sigma_{dB} \sqrt{2\pi}} \int_0^\infty \frac{1}{a} \exp\left[-\frac{(20 \log(a) - M_{dB})^2}{2\Sigma_{dB}^2}\right] \exp\left[-\frac{r^2 + a^2}{2\sigma^2}\right] I_0\left(\frac{ra}{\sigma^2}\right) da \quad (8)$$

where $\sigma = \sqrt{10^{MP/10}/2}$, with MP the normalized average multipath power expressed in dB.

The Loo distribution is very versatile as it includes special cases such the normal and the Rice distributions for large values of a , and the Rayleigh distribution when the value of a is negligible. This property makes this distribution valid for a very wide range of conditions spanning from LOS to heavily shadowed (blockage) conditions [4].

For convenience, we reproduce here the formulation used by Loo in [1] and present its equivalent parameters using the conventions of this paper. Thus, the expression in [1] for the Loo pdf was

$$f(r) = \int_0^\infty f(r|z)f(z) dz = \frac{r}{b_0 \sqrt{2\pi d_0}} \int_0^\infty \frac{1}{z} \exp\left[-\frac{(\ln(z) - \mu)^2}{2d_0}\right] \exp\left[-\frac{r^2 + z^2}{2b_0^2}\right] I_0\left(\frac{rz}{b_0}\right) dz \quad (9)$$

where the direct signal is called z , b_0 is equivalent to σ^2 , μ to M_{NP} and $\sqrt{d_0}$ to Σ_{NP} . Later, in Section 4 where several cases are discussed and simulated, this issue of parameter equivalences is further discussed.

In the case of the C&V [2] model, similar but not identical assumptions are made; i.e. the overall received amplitude, r , over long lengths of the mobile route (and within a given Markov state) is given by

$$f(r) = \int_0^\infty f(r|s)f(s) ds \quad (10)$$

with $r \geq 0$, where $f(r|s)$ is the Rice pdf conditioned on a given shadowing level, s ; thus [2],

$$f(r|s) = 2(k+1) \frac{r}{s^2} \exp\left(-(k+1)\left(\frac{r}{s}\right)^2 - k\right) I_0\left(2\frac{r}{s}\sqrt{k(k+1)}\right) \quad (11)$$

where k is the carrier-to-multipath ratio (in linear units) as defined previously. This form of the Rice distribution results from assumption (b) discussed above.

Variable s describing the shadowing effect is defined by the expression $a^2 + 2\sigma^2 = s^2$ that represents the total normalized power, including the direct and the multipath components. The shadowing variable, s , is assumed to follow a lognormal distribution given by

$$f(s) = \frac{1}{s \Sigma_{NP} \sqrt{2\pi}} \exp\left[-\frac{(\ln(s)M_{NP})^2}{2\Sigma_{NP}^2}\right] = \frac{8.686}{s \Sigma_{dB} \sqrt{2\pi}} \exp\left[-\frac{(20 \log(s)M_{dB})^2}{2\Sigma_{dB}^2}\right] \quad (12)$$

Note that s^2 represents a power, while s represents a voltage, thus the use of $20 \log(s)$ for converting it to dB.

The overall signal envelope can be interpreted as the product of two independent processes, i.e. $r_C \vee = r_{\text{Rice}} s_{\text{LN}}$. In terms of phasors, the overall received complex envelope is given by

$$r_T \exp(j\phi_T) = s[r_D \exp(j\phi_D) + r_M \exp(j\phi_M)] \quad (13)$$

That is, a real, lognormally distributed series, s , is multiplied by a Rice distributed complex phasor.

The essential difference between the Loo model and that of C&V is this assumption of independence. For independent s and r distributions, the following expression is valid [2,10]:

$$f_{C\ V}(r) = \int_0^\infty \frac{1}{s} f_{\text{Rice}}\left(\frac{r}{s}\right) f_{\text{LN}}(s) ds \quad (14)$$

with $r \geq 0$. Comparing (10) and (14), it is possible to arrive at the expression [2]

$$f(r|s) = \frac{1}{s} f_{\text{Rice}}\left(\frac{r}{s}\right) \triangleq \frac{r}{s^2 \sigma_R^2} \exp\left[-\frac{1}{2} \left(\frac{r^2}{s^2 \sigma_R^2} + 2k\right)\right] I_0\left(\frac{r}{s \sigma_R} \sqrt{2k}\right) \quad (15)$$

which implies that $\sigma_R^2 = [2(k+1)]^{-1}$. We can calculate the corresponding CDF using

$$\begin{aligned} \text{CDF}(R) &= \text{Prob}(r \leq R) = \int_0^\infty \int_0^R \frac{1}{s} f_{\text{Rice}}\left(\frac{r}{s}\right) f_{\text{LN}}(s) ds dr \\ &= 1 - E\left[Q\left(\sqrt{2k}, \frac{R}{s} \sqrt{s(k+1)}\right)\right] \end{aligned} \quad (16.a)$$

or, equivalently,

$$\text{CCDF}(R) = \text{Prob}(r > R) = E\left[Q\left(\sqrt{2k}, \frac{R}{s} \sqrt{s(k+1)}\right)\right]. \quad (16.b)$$

where $E[\]$ is the average with respect to s , and Q is Marcum's Q function given by

$$Q(a, b) = \int_a^\infty x \exp\left(-\frac{x^2 + a^2}{2}\right) I_0(ax) dx \quad (17)$$

where a and b are nonnegative real numbers [11].

Again, for reference, the original formulation used in C&V's paper [2] is reproduced here. Thus, the expression for $f(r|s)$ is identical to (11) while the lognormal pdf was in the form

$$f(s) = \frac{1}{\sqrt{2\pi} h \sigma s} \exp\left[-\frac{1}{2} \left(\frac{\ln(s) - \mu}{h\sigma}\right)^2\right] \quad (18)$$

where $h = \ln(10)/20 = 1/8.686$, μ is in Nepers, σ is in dB and parameter k is in linear units.

In the same way as in the Loo case, the C&V distribution encompasses other distributions as special cases: if $k = 0$, i.e. when there is no direct signal present, the Suzuki distribution (described below) results; in case $\Sigma = 0$ (the standard deviation of the lognormal part), the C&V distribution tends to a Rice distribution.

In the *Suzuki* distribution [3] case, similar assumptions are made, i.e. the fast variations due to multipath are Rayleigh distributed, i.e.

$$f(r|\sigma) = \frac{r}{\sigma^2} \exp\left[-\frac{r^2}{2\sigma^2}\right] \quad (19)$$

with $r \geq 0$. The distribution parameter, $2\sigma^2$, remains constant for short sections of the route. For longer routes, lognormal variations of σ can be assumed, i.e.

$$f(\sigma) = \frac{1}{\Sigma_{\text{Np}} \sigma \sqrt{2\pi}} \exp\left[-\frac{(\ln(\sigma) - M_{\text{Np}})^2}{2\Sigma_{\text{Np}}^2}\right] = \frac{8.686}{\Sigma_{\text{dB}} \sigma \sqrt{2\pi}} \exp\left[-\frac{(20 \log(\sigma) - M_{\text{dB}})^2}{2\Sigma_{\text{dB}}^2}\right] \quad (20)$$

The overall Suzuki distribution is given by

$$f(r) = \int_0^\infty f(r|\sigma)f(\sigma) d\sigma = \frac{8.686r}{\Sigma_{dB}\sqrt{2\pi}} \int_0^\infty \frac{1}{\sigma^3} \exp\left[-\frac{(20 \log(\sigma) - M_{dB})^2}{2\Sigma_{dB}^2}\right] \exp\left[-\frac{r^2}{2\sigma^2}\right] d\sigma \quad (21)$$

This distribution, typical in cellular mobile applications, has been included in this paper on the LMS channel given that, as discussed earlier, larger fade margins are expected to be available in new SSP broadcast systems, allowing limited operation within intermediate to heavy shadowing cases. Moreover, a similar distribution, discussed below, was proposed by Lutz *et al.* [12] for the BAD state.

In phasor form, the Suzuki distributed complex envelope consists of a single component, that of the multipath or random component, whose parameter, σ , is slowly changing as the user terminal moves along its route, i.e.

$$r_T \exp(j\phi_T) = \sigma r_M \exp(j\phi_M) \quad (22)$$

Parameter σ is the mode of the Rayleigh distribution, and its corresponding *root mean square* (RMS) value is $\sqrt{2}\sigma^2$. Similar to what happened for the other two distributions, special or extreme cases exist in which the Suzuki distribution matches simpler distributions, e.g. if $\Sigma = 0$ a Rayleigh distribution results.

As just mentioned, in some studies, as in [12], it is not the normalized received amplitude (voltage) that is modeled but the normalized power, p' , where $p' = r^2$. In this case, for heavy-shadowed environments, e.g. urban areas, similar assumptions were made to those of the Suzuki distribution.

To a Rayleigh distribution for the amplitude, r , corresponds an exponential distribution for the power, p' . In this case, the normalized received power distribution is given by

$$f(p'|s_0) = \frac{1}{s_0} \exp\left[-\frac{p'}{s_0}\right] \quad (23)$$

with $p' \geq 0$, and where s_0 is the average (and also the standard deviation) of the distribution. The relationship $s_0 = 2\sigma^2$ is fulfilled; that is, the average of the distribution of the power is equal to the RMS squared value of the distribution for the amplitude.

In addition, s_0 is assumed to be lognormally distributed, i.e.

$$f(s_0) = \frac{1}{\Sigma_{Np}s_0\sqrt{2\pi}} \exp\left[-\frac{(\ln(s_0) - M_{Np})^2}{2\Sigma_{Np}^2}\right] = \frac{4.343}{\Sigma_{dB}s_0\sqrt{2\pi}} \exp\left[-\frac{(10 \log(s_0) - M_{dB})^2}{2\Sigma_{dB}^2}\right] \quad (24)$$

where M_{dB} and Σ_{dB} are the mean and the standard deviation of $10 \log(s_0)$. The overall distribution of the normalized received power variations (within a heavy-shadowed state) is given by

$$f(p) = \int_0^\infty f(p|s_0)f(s_0) ds_0 = \frac{4.383}{\Sigma_{dB}\sqrt{2\pi}} \int_0^\infty \frac{1}{s_0^2} \exp\left\{-\frac{[10 \log(s_0) - M_{dB}]^2}{2\Sigma_{dB}^2}\right\} \exp\left[-\frac{p'}{s_0}\right] ds_0 \quad (25)$$

Note that the basic difference between the parameters of the above distribution and those of Suzuki is that

$$M_{dB}(\text{Suzuki, voltage}) = M_{dB}(\text{Lutz, power}) - 3 \text{ dB} \quad (26)$$

while $\Sigma_{dB}(\text{Suzuki}) = \Sigma_{dB}(\text{Lutz})$. The above relationships are important when trying to use model parameters reported in the literature. It is important to understand the different conventions taken and make the necessary conversions.

Other, more sophisticated models have been proposed, which, instead of two or three parameters, require four or more. An enhancement of the Loo model was proposed in [13], where, as in Loo's model, the direct signal is subjected to lognormal variations, s_1 ; but, in this case, also the multipath component undergoes its own lognormal variations, s_2 . In phasor form, this model follows the expression

$$r_T \exp(j\phi_T) = s_1 r_D \exp(j\phi_D) + s_2 r_M \exp(j\phi_M) \quad (27)$$

An evolution of the C&V model was proposed in [14,15] called the *generalized lognormal* model. In this case, in addition to the Ricean distributed series multiplied by slower, lognormal fluctuations, a new Rayleigh phasor is considered, i.e.

$$r_T \exp(j\phi_T) = s[r_D \exp(j\phi_D) + r_{M1} \exp(j\phi_{M1})] + r_{M2} \exp(j\phi_{M2}) \quad (28)$$

Many other models have been proposed in the literature. In [16] a comprehensive list and a description of statistical models are provided. They all provide reasonable explanations for the time/traveled distance evolution of the received signal. However, from a system analysis point of view, such increased complexity, which is important from the point of view of fully understanding the channel, may not be necessary in most cases.

3. GENERATING SYNTHETIC TIME-SERIES

For simulation purposes, the previously described statistical models translate into circuit form as shown in Figures 4 and 11–13. These circuitual representations not only consider the reproduction of the amplitudes (magnitude of the complex envelope) but also take into consideration the different rates of variation found in real-life conditions whereby multipath fading is *fast varying* while shadowing is *slowly varying*. Additionally, phase and Doppler spectra characteristics are also introduced in the synthesized complex envelopes, $r_T(t) \exp(j\phi_T(t))$.

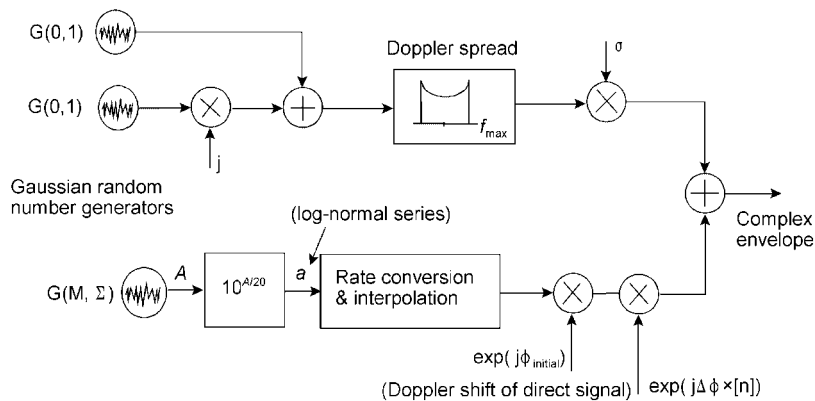


Figure 4. Circuit implementation of the Loo model with Jakes multipath Doppler shaping.

Figure 4 shows the circuit implementing the Loo model: two zero-mean and unit-standard deviation Gaussian series in quadrature are passed through a unit-energy, low-pass, U-shaped filter thus simulating the so-called Jakes spectrum [6] corresponding to an uniform azimuth angle-of-arrival distribution of the multipath contributions. A constant elevation with respect to the receive antenna of 0° is also assumed in the Jakes model. The maximum Doppler spread is $\pm f_{\max} = \pm v/\lambda$, with v the mobile speed. The spectral shaping is described by the expression

$$S_{\text{Doppler}}(f) = \begin{cases} \frac{A}{\sqrt{1-(f/f_{\max})^2}} & \text{for } |f| \leq f_{\max} \\ 0 & \text{for } |f| > f_{\max} \end{cases} \quad (29)$$

where the constant A can be used to force an overall filter energy equal to one so that the standard deviation of the complex Gaussian process (multipath components) is not changed after filtering. Figure 5 shows a simple implementation of the Jakes Doppler filter based on calculating the inverse fast Fourier transform (IFFT) of the filter's frequency response, also shown in the figure.

An alternative, more realistic Doppler power spectrum is described in [6,17]:

$$S_{\text{Doppler}}(f) = \begin{cases} 0 & |f| > f_{\max} \\ \frac{A}{4 \sin \beta_m} \left(\frac{1}{f_{\max}} \right) & f_{\max} \cos \beta_m \leq |f| \leq f_{\max} \\ \frac{A}{f_{\max}} \left\{ \frac{\pi}{2} - \sin^{-1} \left[\frac{2 \cos^2 \beta_m - 1 - (f/f_{\max})^2}{1 - (f/f_{\max})^2} \right] \right\} & |f| < f_{\max} \cos \beta_m \end{cases} \quad (30)$$

which corresponds to the case where the multipath contributions are also spread out in elevation up to a maximum/minimum angle, $\pm \beta_m$.

After Doppler shaping (unit-energy filter), the resulting complex series is multiplied by σ , with $2\sigma^2$ being the normalized average multipath power.

The lower rail in the diagram in Figure 4 simulates the direct signal's amplitude and phase variations. First, an M (dB) mean and Σ (dB) standard deviation Gaussian distributed series is generated. The series, in dB, is converted to linear units (lognormal distributed) by computing its antilogarithm $a = 10^{A/20}$. Then, the phase variations in the direct signal are introduced. These are assumed to be linearly varying giving rise to a constant Doppler spectral line that depends on the mobile velocity and the angle-of-arrival, the azimuth and the elevation, with

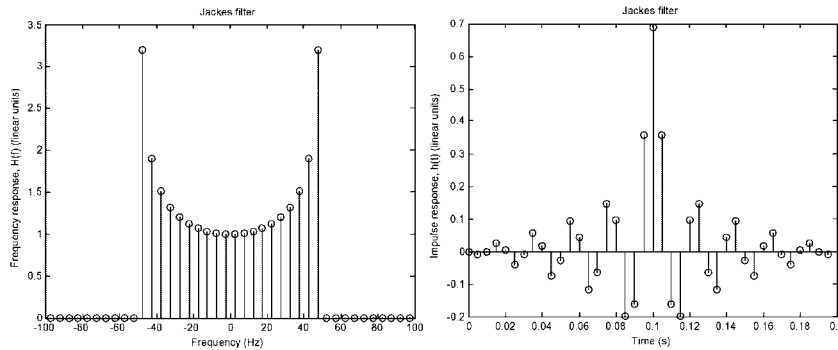


Figure 5. Jakes Doppler filter in the frequency domain and its IFFT.

respect to the mobile trajectory, i.e.

$$f_{\text{Dir}} = (v/\lambda) \cos(\varphi) \cos(\theta) = f_{\text{max}} \cos(\varphi) \cos(\theta) \quad (31)$$

with φ and θ being the relative azimuth (route orientation with respect to the satellite) and elevation angles, respectively (Figure 6). The constant phase increment in Figure 4 is given by

$$\Delta\phi_{\text{Dir}} = 2\pi f_{\text{Dir}} T_s \quad (32)$$

where T_s is the sampling interval in seconds used in the time-series synthesizer. In the simulator block diagram, Figure 4, an initial phase which can be randomly drawn has been included.

An alternative to the Jakes Doppler model is using a Butterworth filter [7] with a transfer function given by

$$|H_{\text{Butt}}(f)|^2 = \frac{A}{1 + (f/f_c)^{2k}} \quad (33)$$

where f_c is the cutoff frequency and k the order of the filter.

Implementing non-symmetric Doppler spectra is also possible by individually filtering the I and Q rails and introducing some degree of cross-talk between the two branches [15,18] (Figure 11).

The above is only valid for short time intervals. It must be born in mind that two Doppler-related phenomena are present, i.e. the Doppler shift and the Doppler spread (Figures 6–8). So far, only the Doppler spread has been discussed.

The Doppler shift is significant in non-GEO satellites where the relative positions of the satellite and the user change. The Doppler shift depends on the variation of the satellite to terminal path length. These shifts can be modeled separately from those due to the movement of the terminal through the multiple scattering environment, the Doppler spread.

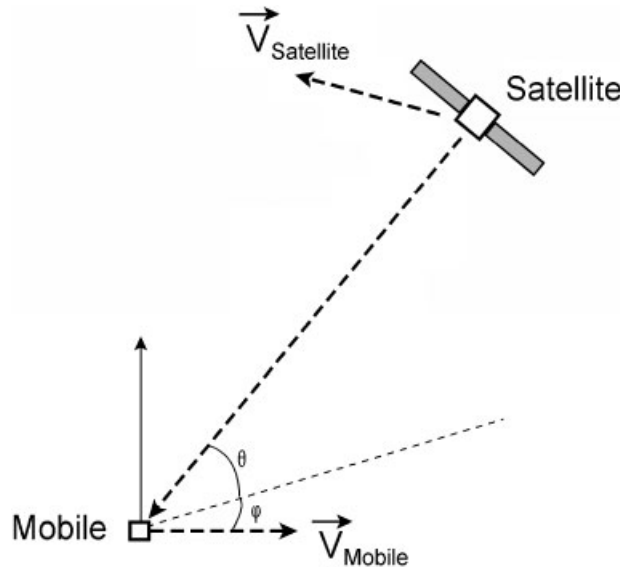


Figure 6. Geometry of the Doppler effects.

Doppler shifts due to satellite motion can be of several kHz although the variation of these shifts is fairly slow, even for LEO satellites at L/S-bands, as illustrated in Figure 7. These variations can be tracked and corrected for at the receiver. Figure 8 illustrates both the Doppler shift and Doppler spread concepts.

Coming back to the variations in the amplitude, the direct signal's amplitude is subjected to variations slower than those due to multipath caused by shadowing. The rate of change of these slow variations is characterized by a correlation time, t_{corr} , or, in the traveled distance domain, the correlation length, l_{corr} (Figure 3). Typically, correlation lengths for these slow variations may be in the order of 3–5 m [19]. The lognormal series generator (lower rail in Figure 4) produces uncorrelated samples that are assumed to be spaced t_{corr} seconds.

At the same time, the fast variations are generated with a much higher sampling rate to account for the bandwidth of the Doppler spread, i.e. $\pm f_{\text{max}} = \pm v/\lambda$. This means that a combined rate conversion plus interpolation block has to be included (Figure 9). The slowly varying direct signal is thus transformed for allowing the complex addition of the direct and the multipath phasors performed at the end of the simulation chain (Figure 4).

As an example, if the mobile terminal travels at a speed, v , of 10 m/s, an L-band carrier frequency of, for example, 1500 MHz is used and if a sampling rate of 8 times (Nf_{max}) the

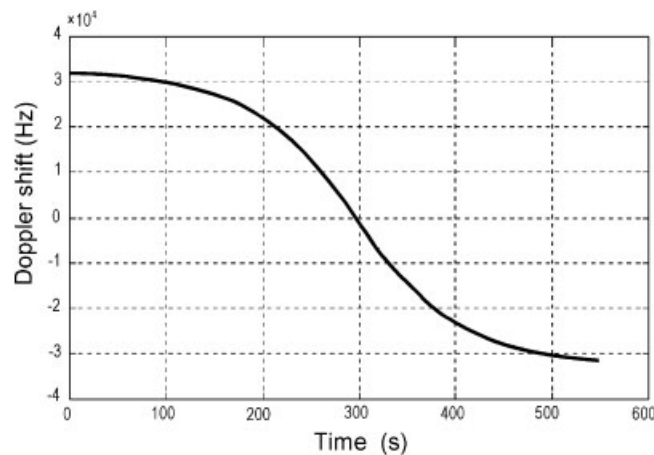


Figure 7. Doppler shift caused by a LEO satellite movement at L-band.

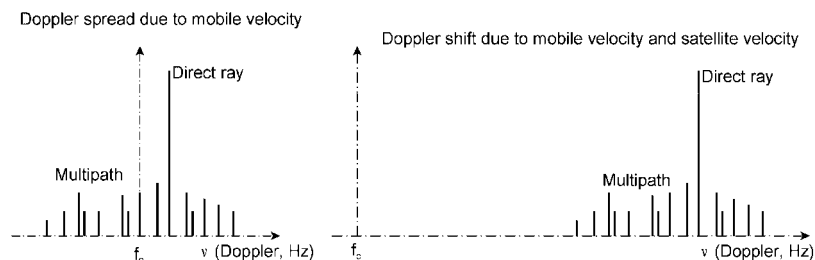


Figure 8. Schematic representation of the Doppler spread and Doppler shift phenomena.

maximum frequency (4 times the Nyquist rate) is employed, the sampling interval would be

$$f_s(\text{fast variations}) = 8v/\lambda = 8 \times 50 = 400 \text{ Hz} \Rightarrow T_s = 0.0025 \text{ s} \quad (34)$$

On the other hand, assuming a correlation length of 3 m (equivalent to a correlation time of 0.3 s for a mobile velocity of 10 m/s) equal to the sampling spacing for the slow variations, a rate conversion of 120 (0.3/0.0025) times is required.

The rate conversion stage is followed by an interpolation process (Figure 9). A spline (cubic) function has been used instead of a linear function. Splines are low-pass filters with very good interpolation properties. Here, no attention has been paid to accurately reproducing the spectral/auto-correlation characteristics of the slow variations. This has been considered acceptable, given the much wider superposed multipath spectrum, to just aim at preserving the value of the correlation time.

An alternative approach could be generating all samples, slow and fast varying, at the same rate ($1/T_s$) corresponding to the sampling rate of the multipath variations. To impose a slower rate on the shadowing variations, a recursive low-pass filter could be used as suggested in [20] (Figure 10). If the correlation length is l_{corr} , the mobile terminal speed, v , the mean, M , and the standard deviation, Σ , then the multiplicative factors in the filter (Figure 10) are given by

$$b = \Sigma \sqrt{1 - c^2} \quad \text{and} \quad c = \exp\left(-\frac{vT_s}{l_{\text{corr}}}\right) \quad (35)$$

The series from the two rails (slow and fast variations), sampled at the same rate, can now be added (complex addition) to produce a Loo distributed time-series that preserves both the rates of change of shadowing and multipath, and the Doppler spectra of both the direct signal and the multipath (Figure 4).

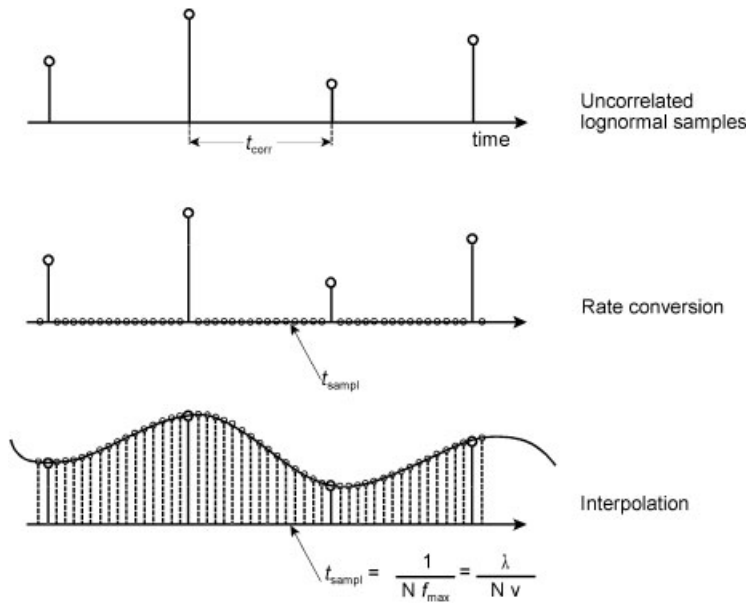


Figure 9. Rate conversion and interpolation of slow variations.

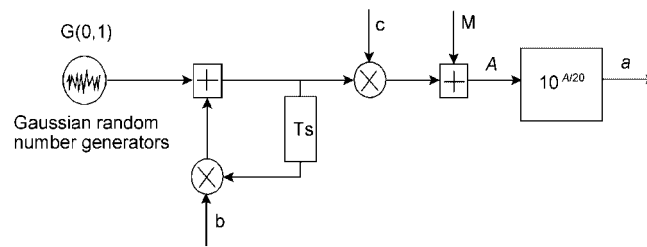


Figure 10. Alternative low-pass filtering method for generating slower varying shadowed signals at the same rate as the multipath rail.

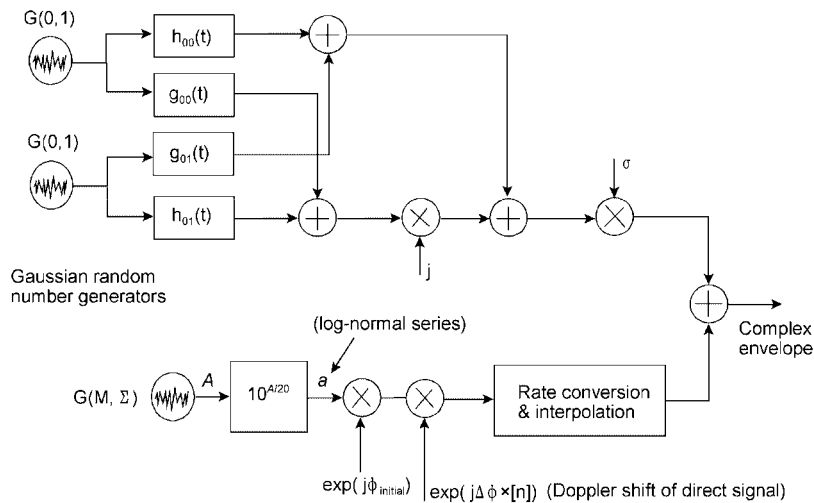


Figure 11. Circuit implementation of the Loo model with asymmetric Doppler shaping.

To implement a simulator for the *C&V* model, a multiplication of a Ricean by a lognormal process must be performed. Many similarities exist between this and the Loo model implementation. In this case, the upper rail in Figure 12 represents the generation of a Ricean time-series with direct signal amplitude $a = \sqrt{k/(k+1)}$ and multipath parameter $\sigma = [2(k+1)]^{-1/2}$. This allows that the overall power be equal to one, i.e. $a^2 + 2\sigma^2 = 1$. In the figure, Jakes multipath Doppler shaping has been assumed.

As for the spectral shaping of the multipath component and the introduction of the direct signal phase, they are implemented in the same way as in the Loo case.

The lower rail in Figure 12 produces the lognormal time-series representing the shadowing process. This series varies at a slower rate than the Rice time-series in the upper rail and, hence, a rate conversion plus interpolation block or a low-pass filter, as in Figure 10, is needed prior to multiplying both series.

Finally, the *Suzuki* simulation circuit is implemented by multiplying a Rayleigh distributed series by a slowly varying lognormal series thus modulating the mode of the Rayleigh series. Then, the Rayleigh series is spectrally shaped. The slow variations are rate converted and interpolated using the same approach as in the two previous cases. For this model, there is no

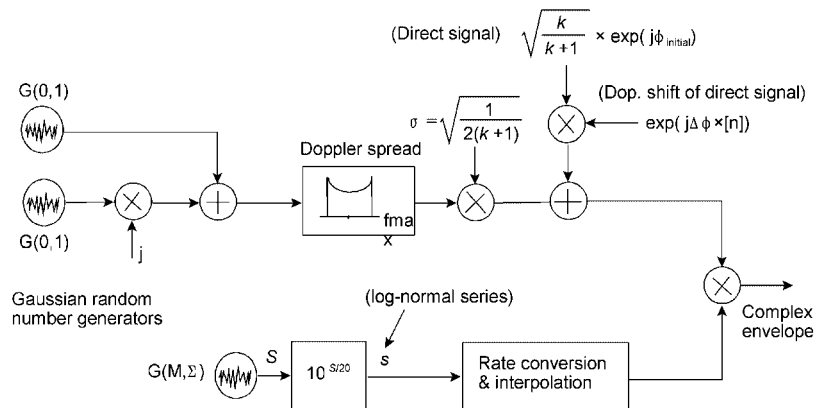


Figure 12. Circuit implementation of the C&V model with Jakes multipath Doppler shaping.

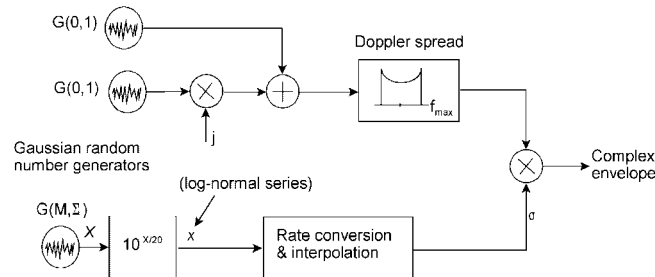


Figure 13. Circuit implementation of the Suzuki model with Jakes multipath Doppler shaping.

Table I. Input parameters in each simulator.

Loo	M_L	Σ_L	MP_L
Corazza and Vatalaro	$M_{C\&V}$	$\Sigma_{C\&V}$	$K_{C\&V}$
Suzuki	M_S	Σ_S	

direct signal involved. The circuit implementation for a Jakes Doppler spectral shaping is shown in Figure 13.

In these three simulators, three Gaussian random generators producing uncorrelated samples have been used. A software implementation has been carried out in MATLAB, which has been used to produce the example results discussed in the following section. The input parameters listed in Table I have been used in each case. The Loo and C&V distributions have three parameters, while the Suzuki distribution has only two, thus being less flexible. As already mentioned in the previous section, the Suzuki distribution is a special case of the C&V distribution. Additional inputs to the simulator are the mobile speed, the correlation time/distance and the number of samples per wavelength.

4. EXAMPLES AND COMPARISONS

In his original paper [1], Loo identified in measured data two extreme behaviors: light and heavy shadowing conditions, which can be described by two different parameter sets. Later, C&V in [2] described the same two cases in terms of their newly defined distribution, thus fitting two parameter sets to the experimental CDFs provided in [1]. In this section, these two cases are simulated and the generated time-series are compared. In addition, Suzuki distributed time-series is also synthesized for comparison, for the heavy shadowing case.

The main interest in this work is verifying the similarities and differences between the generated series, not only in terms of their amplitude CCDFs but also in terms of their phases and Doppler spectra. Thus, we analyzed the two cases identified by Loo: Case 1 corresponding to light shadowing conditions and Case 2 to heavy shadowing. Loo provided the model parameters reproduced in Table II.

There is a slight discrepancy between the parameter definitions in this, and those in Loo's paper. As for the specification of the mean and the standard deviation of the direct signal's amplitude variations, for converting linear units to dB, in this paper $20 \log$ has been used, whereas in Loo's paper $10 \log$ was used. Thus, for making comparisons we have worked out the linear unit values for d_0 and μ and converted them to dB using $20 \log$. The new calculated values are summarized in Table III. Regarding the multipath specification, Loo defines the average multipath power as $10 \log(b_0)$, which is equivalent to $10 \log(\sigma^2)$, whereas, in this paper, the convention is $MP = 10 \log(2\sigma^2)$.

To clarify things, the calculation steps followed for going from the parameter sets in Table II to those in Table III are presented. Thus, the Loo model parameters according to the definitions used in this paper are as follows:

$$M_{dB} = 2 \times 10 \log(\mu), \quad \Sigma_{dB} = 2 \times 10 \log(\sqrt{d_0}) \quad \text{and} \quad MP_{dB} = 10 \log(b_0) + 3 \quad (36)$$

Hence, the new set of Loo model parameters is listed in Table III.

On the other hand, C&V in [2] presented their fitted distribution parameters for Cases 1 and 2. These are listed in Table IV. Again these parameters require their translation according to the

Table II. Summary of Loo model parameters for Cases 1 and 2 as defined in [1].

Conditions	Mean $10 \log(\mu)$ (dB)	Standard deviation $10 \log(\sqrt{d_0})$ (dB)	Multipath power $10 \log(b_0)$ (dB)
Infrequent light shadowing	0.5	0.5	-8.0
Frequent heavy shadowing	-17.0	3.5	-12.0

Table III. Loo model parameters for Cases 1 and 2 using the conventions assumed in this paper.

	M (dB)	Σ (dB)	MP (dB)	Average K (dB)
Light shadowing	1	1	-5	6
Heavy shadowing	-34	7	-9	-25

definitions used in this paper. The conversion process is as follows:

$$M_{\text{dB}} = 8.686\mu, \quad \Sigma_{\text{dB}} = \sigma \quad \text{and} \quad K_{\text{dB}} = 10 \log(k) \quad (37)$$

and these results are summarized in Table V. For Case 1 an alternative value for μ is provided, which achieves a better fit to the original Loo Case 1 CCDF.

Figure 14 shows the two families, Loo and C&V, of CCDFs obtained from the theoretical expressions given in Section 2. The agreement is quite good even though it can be observed how the two distributions are not exactly the same. As said, a closer fit can be achieved if the alternative C&V parameters given in Tables IV and V for the light shadowing case are used.

Next, simulation results corresponding to Case 1, light shadowing, are presented. Figure 15 shows the resulting Loo and C&V magnitude time-series in dB. As the Suzuki distribution assumes that the direct signal is negligible, i.e. $a \approx 0$, this distribution is not applicable to this

Table V. C&V model parameters for Cases 1 and 2 using the conventions of this paper.

	M (dB)	Σ (dB)	K (dB)
Light shadowing	1.13 (1.9978 improved fit)	1	6
Heavy shadowing	-9.38	2.5	-2.22

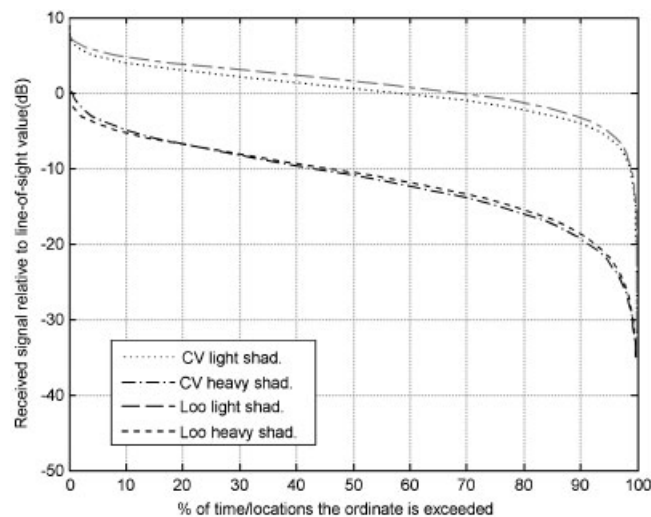


Figure 14. Loo CCDFs for the light and heavy shadowing cases and corresponding C&V CCDFs.

Table IV. Original C&V model parameters for Cases 1 and 2 [2].

	μ (Np)	σ (dB)	k (linear)
Light shadowing	0.13 (0.23 improved fit)	1.0	4.0
Heavy shadowing	-1.08	2.5	0.6

case and no Suzuki series simulation was carried out. Also in Figure 15 the slow variations corresponding to the direct signal (Loo) and the multiplicative shadowing process (C&V) are shown. The same three random generator series were used in all examples for both cases to facilitate the comparisons. The corresponding series CCDFs match the theoretical ones for sufficiently long simulation runs.

In the simulations shown, the abscissas are in traveled distance units (meters). The receiver was assumed to drive in the direction of the satellite that was supposed to have an elevation of 30° . Thus the direct signal's Doppler shift is $(v/\lambda) \cos(\beta) = \frac{10}{0.2} \cos(30) \cos(0) = 43.3$ Hz, where a mobile terminal speed of 10 m/s (36 km/h) has been assumed. The maximum multipath originated Doppler spread is still $\pm v/\lambda$, i.e. ± 50 Hz. To these Dopplers, the Doppler shift caused by the satellite movement has to be added. We assumed that this satellite contribution was zero. The correlation length was set to 9 m for clearer plots, while the number of samples per wavelength was set to 4.

From a detailed inspection of Figure 15 no clear difference may be observed. More information can neither be drawn from Figure 16 where the resulting absolute phases of the Loo and C&V series are plotted. Figure 17 illustrates the resulting Doppler spectra. It can be concluded that, apart from very slight differences in the CCDFs, specially at the tails, the two series are practically identical.

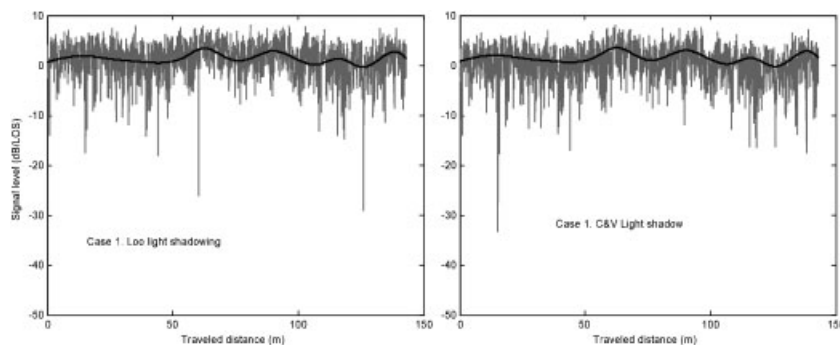


Figure 15. Case 1, light shadowing, time-series for Loo and C&V cases: slow and overall variations in dB.

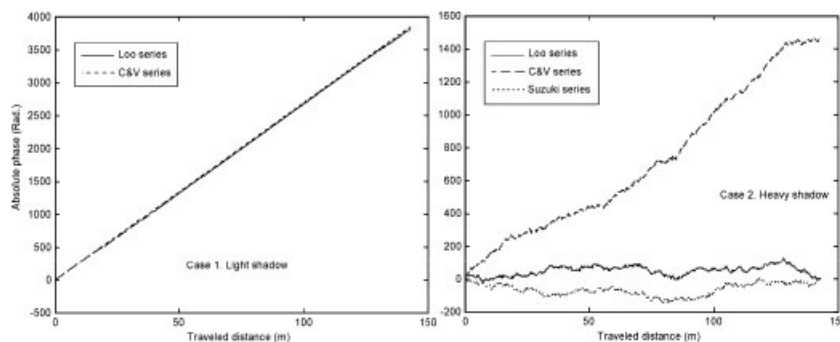


Figure 16. Absolute (unwrapped) phases for Case 1 (light shadowing) and Case 2 (heavy shadowing).

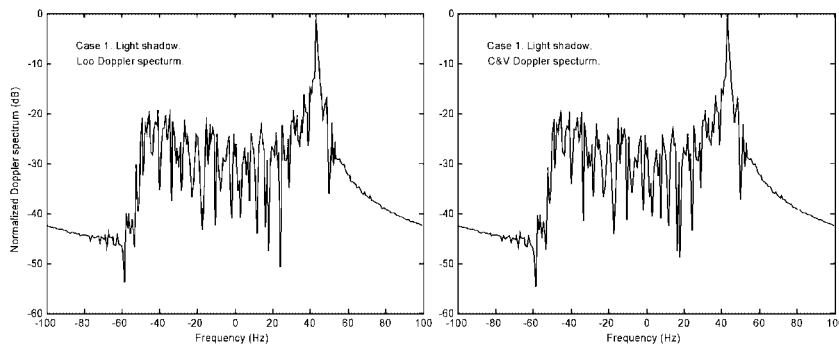


Figure 17. Case 1 Doppler spectra of Loo and C&V time-series.

Things change quite significantly for Case 2, heavy shadowing. Figure 18 illustrates the resulting Loo, C&V and Suzuki amplitude series in dB. Now the series look different even though there is a good match between their respective CCDFs. We discuss the Suzuki CCDF later below. The differences appear in the case of the slow variations, but they need not be identical as they represent different things: the direct signal (Loo), the shadowing effect (C&V) or the variations in the mode of the Rayleigh distribution, in the Suzuki case.

More information is provided by the phase series in Figure 16. It is clear that the C&V series shows an increasing phase due to the direct signal, only distorted by the multipath-induced phase variations present in the Rice process. However, the Loo series is fairly random, meaning that as the direct signal is very much attenuated, the phase of the multipath component dominates. The phase variations in the Suzuki series should be identical to those in the Loo series, as the same three random number series have been used throughout. The difference comes when not so attenuated values of the direct ray take place; in those cases, the phase tends to approach the same linear trend as in the C&V case.

Confirmation of the above observations can be found in Figure 19 where the respective resulting Doppler spectra are shown. In the C&V case, the U-shaped Jakes spectrum is accompanied by a clear delta corresponding to the direct signal, present all the time. In the Loo case, the direct signal is less significant as it only has an influence during the periods when it is not totally attenuated. In the Suzuki case, the direct ray contribution is altogether absent.

A short study has also been performed for the Suzuki distribution in comparison with the two shadowing cases identified by Loo. A very good fit to Case 2 has been achieved with parameters $M = -12$ dB and $\Sigma = 1$ dB (series in Figure 18). A scan of parameter M from -2 to -17 dB and Σ equal to 1 and 2 dB is shown in Figure 20. From this figure it is possible to see that light shadowing conditions are difficult to reproduce with this distribution as the deep fades for probabilities of 90% or higher are difficult to reproduce without generating excessively high signal enhancements for probability levels of 10% or lower. As for the heavy shadowing cases, they are easily reproduced with this distribution.

From the considerations above, the following observations can be made:

- Basically all three models can approximately reproduce the same cumulative distributions, except for the Suzuki model that fails in light shadowing conditions.

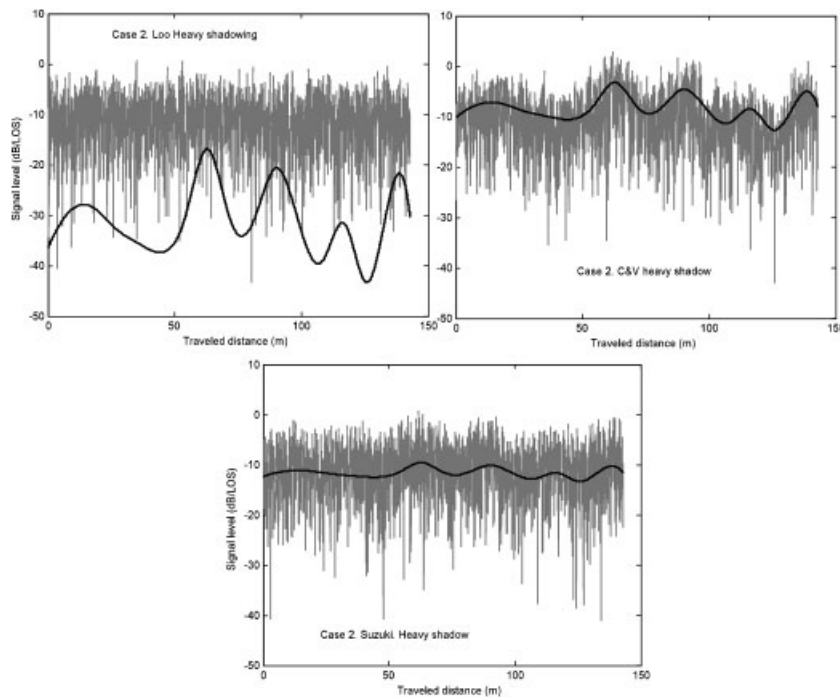


Figure 18. Case 2 Loo, C&V and Suzuki time-series: slow and overall variations in dB.

- However, when observing the local behavior of the resulting fading amplitude series, the C&V model produces time-series that, always, present a Ricean character, while the Suzuki model always produces series with a Rayleigh character. In the Loo model case, its Rice/Rayleigh character depends on how attenuated the lognormal distributed direct signal is locally.
- The above observations are confirmed when looking at the phase and Doppler spectra of the synthetic series corresponding to the three models. It can be observed how, in the C&V case, the phase is almost linear with some superposed fluctuations. This corresponds to typical Ricean conditions. In the Suzuki case, the phase is randomly varying with no dominant trend as it should be for Rayleigh fading. For the Loo model, during the deep fades the Rayleigh character dominates, while, during the less severe fades, the overall phase reflects the variations of the direct signal.
- Of course, the C&V and Loo models are more versatile than the Suzuki model given that this has only two parameters while the others have three parameters. In another section, models with more than three parameters have been briefly discussed. In the authors' opinion, if more complex models are needed, it is more flexible to incorporate statistical models such as those due to Loo, C&V or Suzuki into a Markov state-based framework than using four or more parameter distributions.

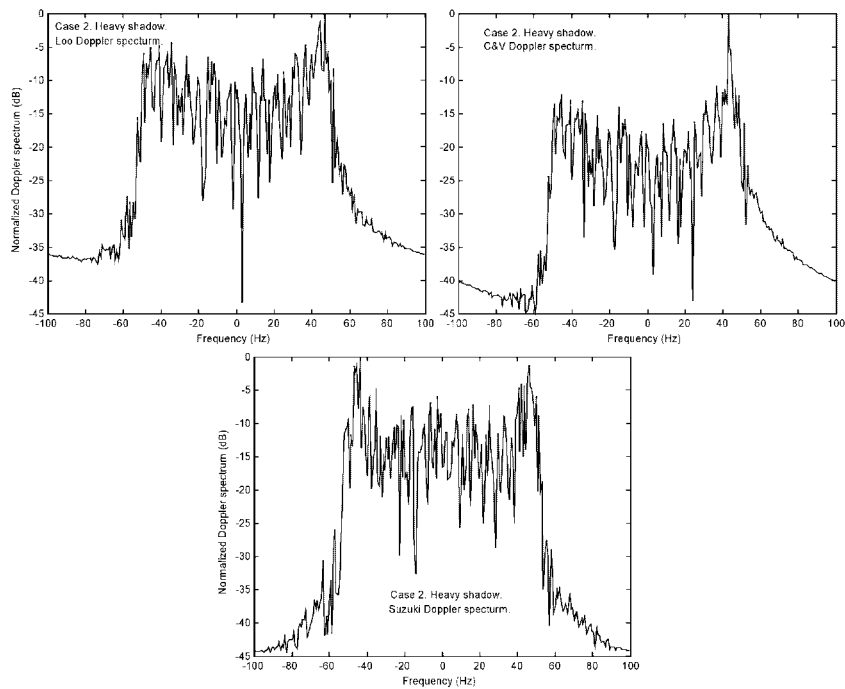


Figure 19. Case 2 Loo, C&V and Suzuki Doppler spectra.

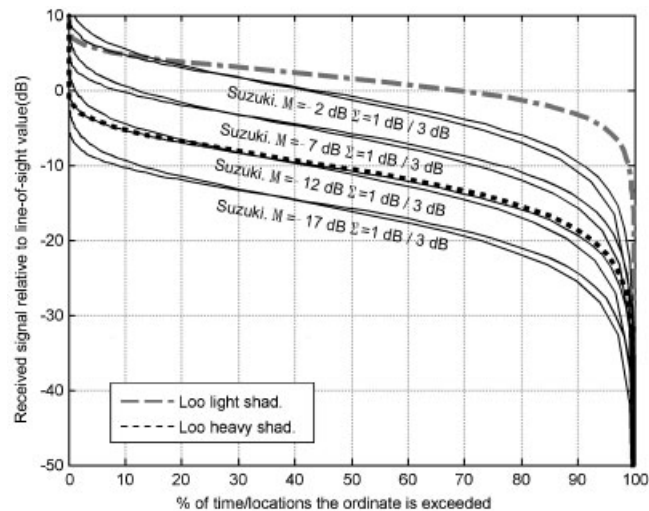


Figure 20. Scan of Suzuki parameters and comparison with Loo's Cases 1 and 2 CCDFs.

5. SUMMARY

In this paper, the assumptions in three statistical models (Loo [1], C&V [2] and Suzuki [3]) used for generating narrowband complex-envelope time-series have been reviewed. In the literature these models have been shown to provide appropriate fits to measured amplitude data. It has been shown that, in most cases, model parameters can be found to produce almost identical cumulative distributions for different channel conditions, e.g. light shadowing, heavy shadowing.

However, when synthetic time-series are produced using the various model assumptions, it can be found that the amplitude series are locally different in that their character is either more Rice-like or Rayleigh-like. These differences can also be found when analyzing the corresponding phase series.

It is not the intention of the authors to indicate which is best, but only to point out their similarities and differences. The authors have used in their earlier work the Loo model [1] as the basis for their modeling work [4,5,19,21].

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AUTHORS' BIOGRAPHIES



Fernando Pérez-Fontán was born in Villagarcía de Arosa, Spain. He obtained his degree in Telecommunications Engineering in 1982 from the Technical University of Madrid and his PhD in 1992 from the same university. After working in industry since 1984 he became an assistant professor at the University of Vigo in 1988. In 1993 he became an associate professor and in 1999 full professor at the Signal Theory and Communications Department of the University of Vigo. He lectures in Radiocommunication Systems, especially in terrestrial fixed and mobile system related topics. He is the author of a number of books and journal papers and has been the leader in several projects funded by public and private entities. He participates in ITU-R WG3 on Propagation Modeling. He has been Management Committee Member of EU' COST 255 and 280, and currently participates with the same role in COST 297.



Antía Mayo Bazarra Telecommunication Engineer, University of Vigo, Spain. During her degree she was carrying out research work on channel simulation, multipath and airborne communication systems. Antía is now working as an IT consultant in Madrid, Spain.



David Marote Alvarez Telecommunication Engineer, University of Vigo, Spain. In 2006 he was granted by ESA-ESTEC to work on Galileo propagation aspects as well as in Ka band reconfigurable antennas. Currently he is working as RF Engineer in EADS CASA Espacio, Madrid.



Roberto Prieto-Cerdeira was born in Ourense (Spain) in 1978. He graduated as a Telecommunications Engineer from the University of Vigo, Spain in 2002. After one year working in the IT industry, he followed postgraduate studies on Space Science and Radioastronomy in Chalmers University of Technology (Gothenburg, Sweden). Since 2004, he has been with the European Space Agency (ESA/ESTEC) in the Wave Interaction and Propagation Section where he is responsible of radiowave propagation activities for ionospheric and local environment effects for satellite communications and for Galileo and EGNOS satellite navigation projects.



Perfecto Mariño received the Ph.D degree in Telecommunications Engineering from the Polytechnic University of Madrid, Spain, in 1984. From 1978 to 1993 he was Associate Professor in the Electronic Technology Department, University of Vigo, Spain. In 1988 he was a Visiting Scientist in the Computer Science Department, Carnegie Mellon University, Pittsburgh, PA, USA. Since 1993 he has been a Full Professor in the Electronic Technology Department, University of Vigo. His main research interests include distributed electronic instrumentation, wireless sensor networks, fieldbus technology for industrial control and artificial vision. He has authored more than 20 articles in national and international technical journals, coauthored 12 books and published more than 100 papers in international conferences. He was technical co-chair for international conferences as IECON, ISIE, SIECI, CAIP, CONTROLLO, SAAEI, and evaluator of several journals such as Mechatronics, IEEE Industrial Electronics, IEEE Industrial Informatics, CIT, ESIME, Elsevier and others. He was involved in more than 40 research and applied projects for several national enterprises as CITROEN, TELEVES, ENCE, UNION FENOSA, ANFACO, CASTROSUA and others. He holds an european patent on distributed instrumentation for power transformers (2005). He is also the Director of the Digital Communications Division from the Electronic Technology Department, University of Vigo. He has been distinguished as an Expert on Information Technology from the Commission of the European Union for the SPRINT (Luxembourg 1991), COPERNICUS (Brussels 1994) and ACTS (Brussels 2002) programs. He has served as General Manager of Information Society (2002–2004) and Communications Technology (2002–2006) programs from the State R&D Office (Autonomous Government, Galicia, Spain). Also is an expert from R&D National Program (Ministry of Education and Science, Madrid) in the evaluation of research projects in the field of information and communication technologies. Senior member of IEEE.



Fernando Machado obtained his M.S. degree in Telecommunications Engineering in 2000 from the University of Vigo, Vigo, Spain, and his PhD in 2008 from the same university. Since 2001, he has been an Assistant Professor in the Electronic Technology Department, University of Vigo. He is currently a researcher with the Digital Communications Division, Electronic Technology Department, University of Vigo. His main research interests include distributed electronic instrumentation, multisensor networks, wireless communications and radiocommunication systems.

Núria Riera Diaz graduated in telecommunications engineering from Polytechnic University of Catalonia (UPC) in 2001. From 2001 to 2008, she has been with DLR as scientific researcher in the Institute of Communications and Navigation, working in Aeronautics and Space as well as IST projects. Since 2008, she is with TriaGnoSys as a senior researcher and system engineer. Her research interests and expertise are in the field of aeronautical communications, satellite systems for air traffic management, system and protocol architectures, mobile personal communications and propagation, in particular ray tracing, and narrow-and wideband statistical modeling. She has been involved in RTCA and EUROCAE working groups.